

CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 05



By - Abhishek Sir



Recap of Previous Lecture



Topic

CRC



Topics to be Covered



Topic

CRC

Topic

2D Parity

ABOUT ME



Hello, I'm **Abhishek**

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- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : CRC



CASE II : Error Included

Receiver protocol :

$$[E(X)] \text{ [Modulo-2 Division] } [G(X)]$$

- $G(X)$ have $(n+1)$ terms (length)
- If $E(X)$ causes at-most n length burst error
then above equation always lead non-zero remainder



Topic : CRC Property ✓



→ CRC can detect any length burst error,
up-to the degree of generator polynomial function ✓



Topic : CRC Property



→ For odd number of errors,
 $E(X)$ must have odd number of terms

→ if $(X+1)$ is a factor of $G(X)$,
then CRC can detect "all odd number of errors"
(all odd bits Error) ✓✓

$E(X)$ modulo-2 division $G(x)$



Topic : CRC Property



→ “ $(X+1)$ is a factor of $G(X)$ ”
means “ $G(X)$ is completely divisible by $(X+1)$ ”
and “ $G(X)$ has even number of terms”

$$\underline{G(X)} = \underline{F_1(X)} * \underline{F_2(X)}$$

$$G(X) = \underline{(X+1)} * F_2(X)$$



Topic : CRC



Example 5 :

$$\underline{G(X)} = \underline{X^3 + 1}$$

$$\text{Transmitted Data} = \boxed{10011101100}$$

 ↑ ↑ ↑

$$E(X) = X^7 + X^5 + X^4$$

$$\text{Received Data} = \boxed{10001011100}$$

#Q. Let $G(X)$ be the generator polynomial used for CRC checking. What is the condition that should be satisfied by $G(X)$ to detect odd number of bits in error?

[GATE 2009]

IIT-R

- ☒ (A) $G(X)$ contains more than two terms
- ☒ (B) $G(X)$ does not divide $1 + X^k$, for any k not exceeding the frame length
- ☐ (C) $(1 + X)$ is a factor of $G(X)$
- ☒ (D) $G(x)$ has an odd number of terms

Ans: C



Topic : CRC Property



→ For two-bit error,
 $E(X)$ must have two terms only

$$\underline{E(X)} = (X^i + X^j)$$

[Where i & j are 0 to $(m + n - 1)$ and $(i > j)$]

$$E(X) = X^j (X^{(i-j)} + 1)$$



Topic : CRC Property



$$E(X) = X^j (X^{(i-j)} + 1)$$

Suppose $k = (i-j)$

$$= X^j * (X^k + 1)$$

→ To detect any two-bit error,

G(X) does not divide $(X^k + 1)$

[for any k from 0 to $(m + n - 1)$]





Topic : CRC

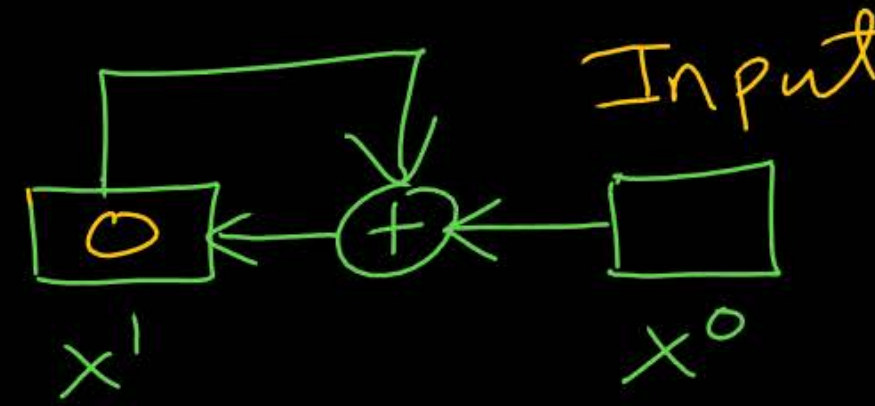


Example 6 :

$$G(X) = X + 1$$

Transmitted DATA = 1 0 1 1 1 0 1 1

One-bit CRC [CRC - 1] is same as one-bit parity with "even parity".





Topic : CRC



Example 7 :

$$G(X) = X^3 + 1$$

Message (DATA) = 1 0 0 1 1 0 0 1

CRC = 0 0 0

$$\begin{array}{r} 1001 \overline{) 1001 \mid 1001 \mid 0000} \\ \underline{1001} \\ 0000 \\ \underline{0000} \\ 0000 \\ \underline{0000} \\ 0000 \end{array}$$

CRC



Topic : CRC



Example 8 :

$$G(X) = X^3 + X^2 + X + 1$$

Message (DATA) = 1 1 1 1 0 0 0 1

CRC = | | |

$$\begin{array}{r}
 1111 \overline{) 111100010000} \\
 \underline{1111} \\
 0000 \\
 \underline{0000} \\
 0000 \\
 \underline{0000} \\
 0000 \\
 \underline{0000} \\
 0000
 \end{array}$$

CRC



Topic : CRC Property



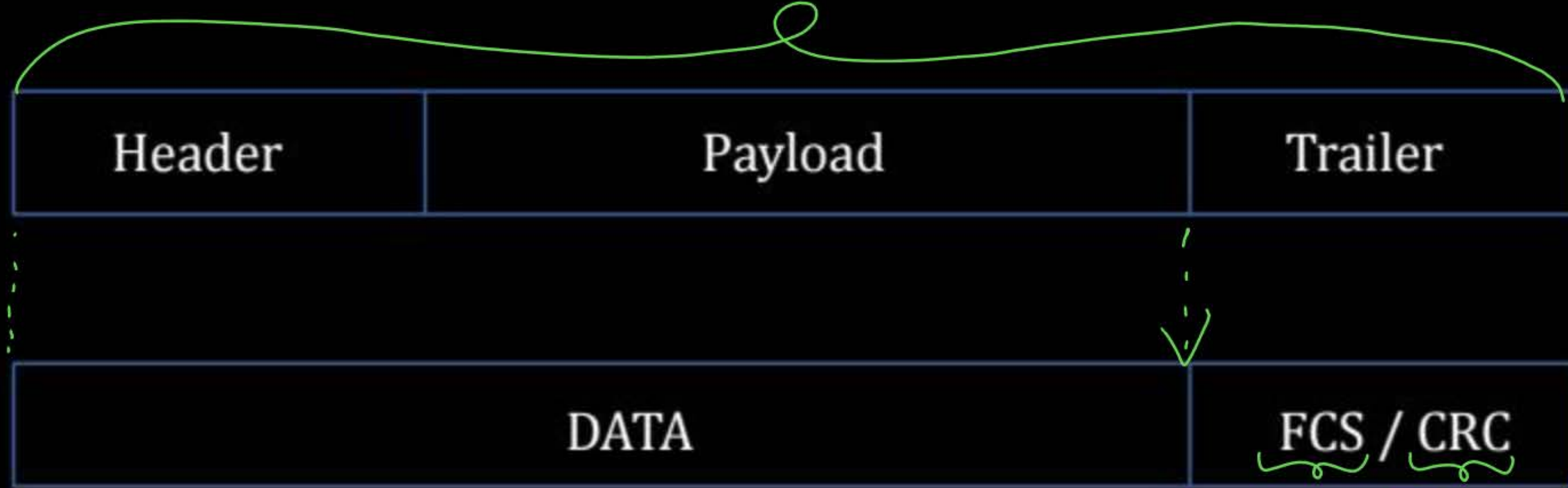
→ CRC can be “all zero bits” and can be “all one bits” ✓



Topic : CRC



Data Link Layer "Frame"



32 bits or (4 bytes)

CRC-32 : [32-bit / 4 byte]

$$G(X) = X^{32} + \dots + 1$$



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A 4x6 grid of binary digits (0s and 1s) with some digits highlighted in orange, green, and white.

0	1	1	0	1	1
1	1	0	1	1	0
1	0	1	1	0	1
0	1	1	0	1	1

The codewords generated by CRC error detection method are 'Cyclic Code'.



Topic : Valid Codewords vs Invalid Codewords



k -bits input (data) $\rightarrow N$ bits output (codeword)

$$\deg_e[G(x)] = n$$

$\rightarrow$ Number of parity bits = ~~$(n - k)$~~ $(N - k) = \underline{n}$

$\rightarrow$ Number of valid codewords = $\underbrace{2^k}$

$\rightarrow$ Number of invalid codewords = ~~$(2^n - 2^k)$~~ = $(2^N - 2^k)$



Topic : Checksum



- Error detection technique *only*.
- Both sender and receiver must agree on same block size (n)
- n bit one's complement of the one's complement sum of all n bit words



Topic : Checksum



- Checksum detect “all single bit error”
- In case of burst error,
checksum able to detect “all odd number of errors”
- Checksum can be “all zero bits”,
but checksum can never be “all one bits”



Topic : Error Detection and Correction

Types of error correction techniques :

1. 2D Parity
2. Hamming Code



Topic : 2D Parity



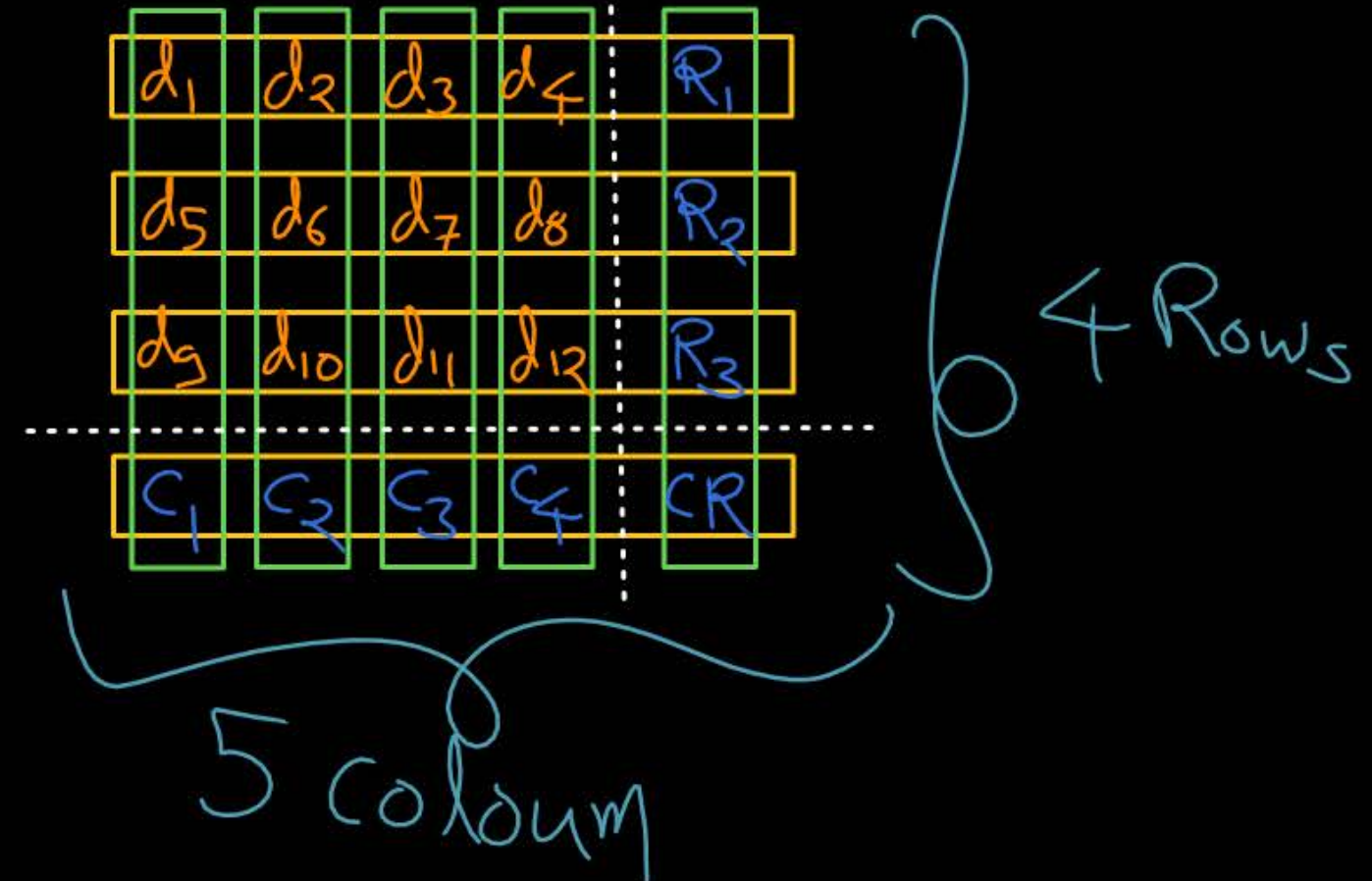
→ Both sender and receiver must agree on same block size and same parity (either even or odd parity)

→ Block Size = $m * n$
[m rows & n columns]

Number of Data bits = $(m-1) * (n-1)$

Number of parity bits = $[m + n - 1]$

[4x5 Block]



CR $\rightarrow$ parity of parity
 [Even Parity]

$$R_1 = d_1 \oplus d_2 \oplus d_3 \oplus d_4$$

$$R_2 = d_5 \oplus d_6 \oplus d_7 \oplus d_8$$

$$R_3 = d_9 \oplus d_{10} \oplus d_{11} \oplus d_{12}$$

$$C_1 = d_1 \oplus d_5 \oplus d_9$$

$$C_2 = d_2 \oplus d_6 \oplus d_{10}$$

$$C_3 = d_3 \oplus d_7 \oplus d_{11}$$

$$C_4 = d_4 \oplus d_8 \oplus d_{12}$$

$$CR = R_1 \oplus R_2 \oplus R_3 = C_1 \oplus C_2 \oplus C_3 \oplus C_4$$



Topic : 2D Parity



Suppose Block Size = 4 x 6 and "Even Parity"

Data = "1 0 0 1 1 0 1 1 0 1 1 0 0 1 0"

Sender

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
0	1	1	0	0	0



Topic : 2D Parity



Suppose **Block Size** = **4 x 6** and “**Even Parity**”

Data = “1 0 0 1 1 0 1 1 0 1 1 0 0 1 0”

Sender

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
					0
0	1	1	0	0	0

Transmitted Data = “1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 0 0 1 1 0 0 0”



Topic : 2D Parity



Suppose **Block Size** = **4 x 6** and "**Even Parity**"

Data = "1 0 0 1 1 0 1 1 0 1 1 0 0 1 0"

CASE I: *No any Error*

Receiver

1	0	0	1	1	1	✓
0	1	1	0	1	1	✓
1	0	0	1	0	0	✓
0	1	1	0	0	0	✓

Sender

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
0	1	1	0	0	0

Transmitted Data = "1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0"

Received Data = "1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0"



Topic : 2D Parity



if receiver finds all row wise and column wise parities are balanced
then receiver concluded "No any error detected"

else

receiver concluded "Error detected"

When Receiver finds any of the parity
is unbalanced
it immediately declare Error Detected.



Topic : 2D Parity



Suppose **Block Size** = **4 x 6** and “**Even Parity**”

Data = “1 0 0 1 1 0 1 1 0 1 1 0 0 1 0”

CASE II: *single bit Error*

Receiver

1	0	0	1	1	1	✓
0	1	0	0	1	1	←
1	0	0	1	0	0	✓
0	1	1	0	0	0	✓
✓	✓	↑	✓	✓	✓	

Sender

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
0	1	1	0	0	0

Transmitted Data = “1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 0 0 1 1 0 0 0”

Received Data = “1 0 0 1 1 1 0 1 0 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0”

[2D parity successfully detect and correct all single bit error.]



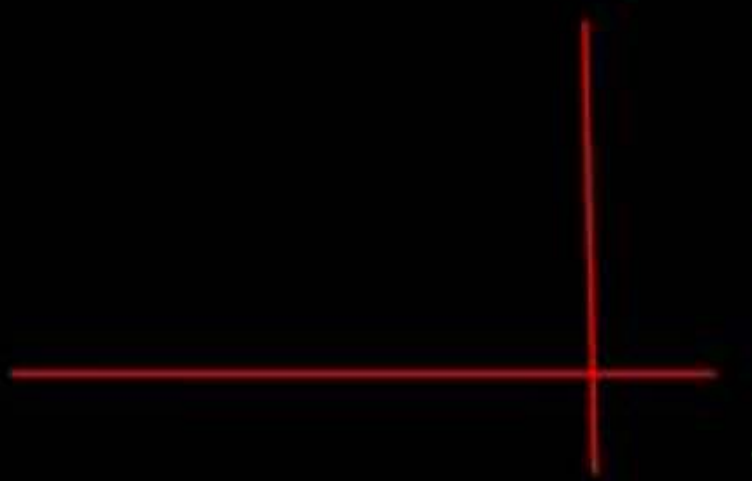
2D → Single bit Error Correction
Parity



Topic : 2D Parity



CASE I : No any error

 } All row wise
Parity are
balanced

All column wise
Parity are balanced.



2 mins Summary



Topic

CRC ✓

Topic

2D Parity



THANK - YOU

